Freight Rate Prediction - Approach, Validation and December Forecast

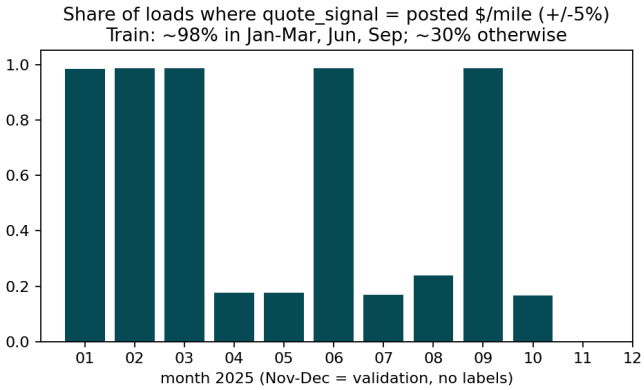
Machine Learning Engineer assessment. Code: see GitHub repository (README.md for run instructions).

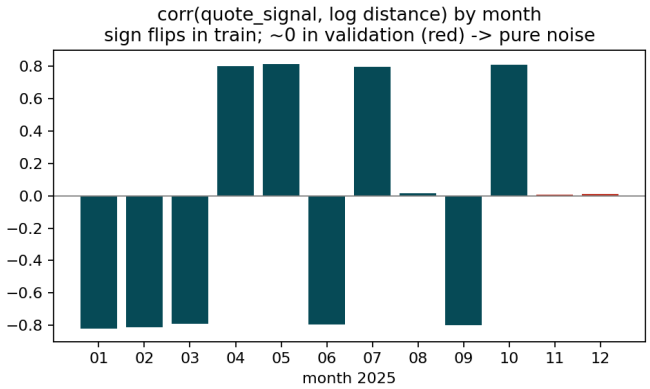
1. Summary

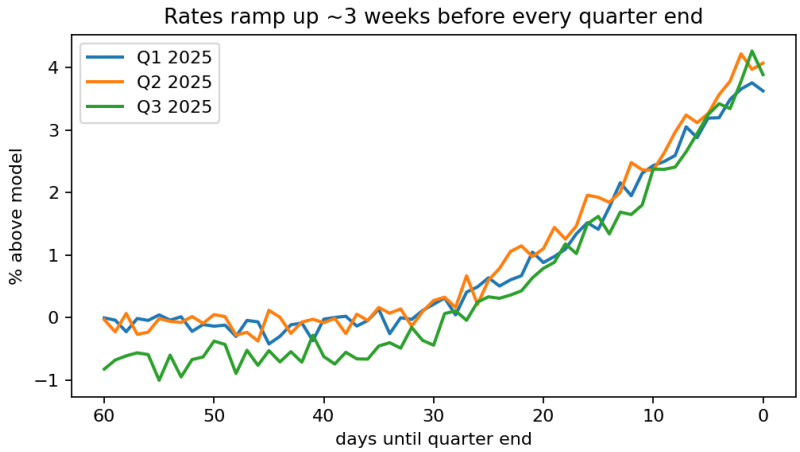
Task: predict **posted_rate** for 12,000 loads dated Nov-Dec 2025 using 48,000 labeled loads from Jan-Oct 2025. Because validation lies entirely *after* training, this is a two-month-ahead forecast, and the data-generating process drifts over time. The result is driven by data understanding rather than model size: on out-of-time folds the final model reaches **1.4% median error / 1.7% MAPE on clean labels** (4.1% MAPE on raw labels, floored by ~1.4% corrupted labels that no model can predict), versus 3.0% / 3.3% for a plain LightGBM and 4.3% / 4.7% for a $/mile table. The estimated noise floor is about 1.8% MAPE.

2. Data exploration and data-quality issues

- **quote_signal is a trap.** In Jan-Mar, Jun and Sep it equals posted $/mile to within ~1% (it would be a near-perfect leak). In Apr-May, Jul-Aug and Oct it is unrelated or inversely related to the rate. In validation it is pure noise: no relation with distance, equipment or the previous day (day-to-day autocorrelation 0.02 vs 0.93 in train). A model trusting it looks great in some CV folds and fails on the final data. **Decision: excluded.** (The 'LightGBM + quote_signal' row below is that experiment: it is worse, not better.)
- **~1.4% corrupted labels (677 rows).** Rates are off by a factor of 0.2-0.3 or 3-5.5 (random up/down) with no relation to any feature; the residual histogram has an empty gap between normal rows (|log error| < 0.3) and these rows (> 0.7). They are flagged by a first-pass fit and excluded from training (without this step the linear component degrades from 1.7% to 7.3% clean MAPE). They still count in the RAW metrics below.
- **Weight:** 292 negative values (sign errors: |w| matches the positive distribution and the rate effect), a hard cap at 47,500 lb (1,204 rows), 300 missing. Handled with abs(), a cap flag and a missing flag; the rate rises ~3% per 10,000 lb.
- **market_index:** noisy per row and 0.8% missing. Replaced by a robust *daily median* (built from features only, so no label leakage); the daily level drives the rate (+0.14 log-points per index point).
- **Distance:** each row carries ~2% noise around its lane's typical distance, and the rate follows the *row* distance (using the lane median instead raised residual noise by ~30%: sd 0.030 vs 0.0225); 69 rows are clipped at a 70-mile floor. City coordinates are one fixed (synthetic/distorted) pair per city; haversine vs distance is erratic for some lanes, so distance is trusted and coordinates are used only as location features.
- **Unseen cities:** validation contains 8 cities absent from train (Chicago, Charlotte, Jackson, Norfolk, Laredo, Allentown, Knoxville, San Diego; 12% of loads). The origin and destination premium of a city are almost identical (corr 0.99) and fall with latitude (south-central cities +6%), so the premium of an unseen city is interpolated from nearby cities' coordinates.
- **Calendar structure.** Rates ramp up over the last ~30 days of every quarter, peaking ~+4% at quarter end - identical in Q1, Q2, Q3 (figure below). Modeled with days-to-quarter-end (transfers to Q4/December; month dummies would not). A trend of ~+0.02%/day and a small day-of-week effect are also present.
- No duplicates or missing ids; Nov-Dec market_index (0.93) lies within the training range.


Share of loads where quote_signal = posted $/mile (+/-5%)
Train: ~98% in Jan-Mar, Jun, Sep; ~30% otherwise
1.0
0.8
0.6
0.4
0.2
0.0
01
02
03
04
05
06
07
08
09
10
11
12
month 2025 (Nov-Dec = validation, no labels)


corr(quote_signal, log distance) by month
sign flips in train; ~0 in validation (red) -> pure noise
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
01
02
03
04
05
06
07
08
09
10
11
12
month 2025


Rates ramp up ~3 weeks before every quarter end
Q1 2025
Q2 2025
Q3 2025
% above model
4
3
2
1
0
−1
60
50
40
30
20
10
0
days until quarter end

3. How the data was split and validated

A random K-fold would place neighbouring days in both train and test, hide the drift, and reward the leaky quote_signal. Instead the validation mimics the real task:

- **Rolling-origin (expanding-window) folds:** train on all data before a cut-off, test on the next two months (same horizon as Nov-Dec): Mar-Apr, May-Jun, Jul-Aug, Sep-Oct. Daily market-index aggregates are label-free and are the only information shared across the boundary.
- **Unseen-city holdout:** 8 random cities (5 draws) are removed from training with every lane touching them, then scored - mirrors the 8 validation-only cities.
- **Two views of each metric:** RAW labels (what an external scorer sees) and CLEAN labels (corrupted rows removed, *same rows for every model*). The metric used by the grader is not published, so MAE, MAPE and MdAPE are all reported. Predictions are the conditional median in log space, which is MAE-optimal and close to MAPE-optimal.
- **Final fit** uses all 48,000 labeled loads; the last fold (Sep-Oct) is the best proxy of Nov-Dec performance. Hyper-parameters barely matter (plateau between 150 and 600 trees), so there is little tuning-on-test risk.

Rolling-origin results (mean of 4 folds)

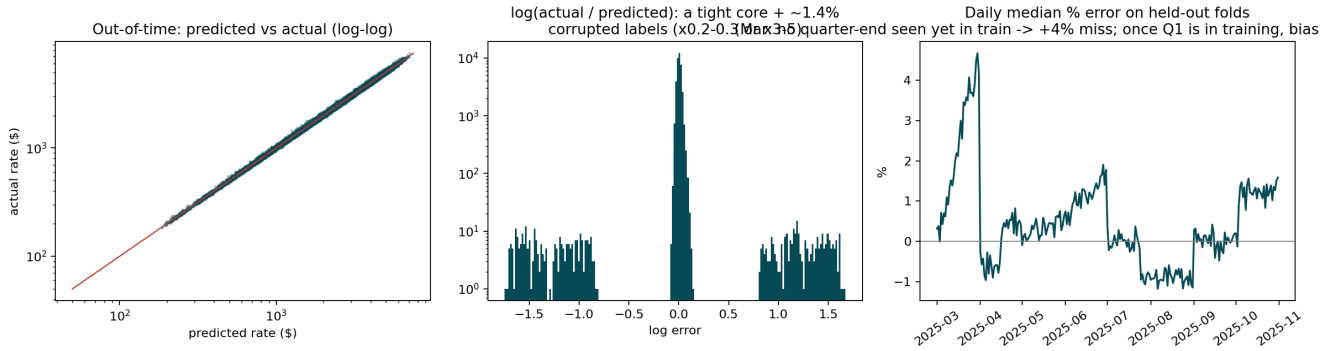
Model	MAE $	MAPE %	MdAPE %	MAE $ (clean)	MAPE % (clean)	MdAPE % (clean)
Naive $/mile table	158.64	7.02	4.33	108.29	4.72	4.27
LightGBM, raw columns	129.54	5.6	2.96	78.89	3.3	2.93
LightGBM + quote_signal (trap)	134.37	5.88	2.75	83.88	3.58	2.7
Additive model, no GBM	95.09	4.2	1.53	43.97	1.82	1.5
Additive model + GBM (FINAL)	**92.13**	**4.09**	**1.44**	**40.97**	**1.71**	**1.41**

Final model by fold

Test fold (train = all earlier data)	Loads	MAE $	MAPE %	MAE $ (clean)	MAPE % (clean)	MdAPE % (clean)
Mar-Apr	9855	94.32	4.56	51.79	2.13	1.67
May-Jun	9696	91.81	3.83	37.96	1.56	1.32
Jul-Aug	9671	87.5	3.8	35.3	1.51	1.26
Sep-Oct	**9523**	**94.9**	**4.16**	**38.82**	**1.64**	**1.41**

Unseen-city holdout (clean labels)

Model	MAE $	MAPE %	MdAPE %
Additive model, no GBM	43.68	1.68	1.4
Additive model + GBM (FINAL)	**40.52**	**1.55**	**1.29**
LightGBM, raw columns	65.65	2.64	2.17

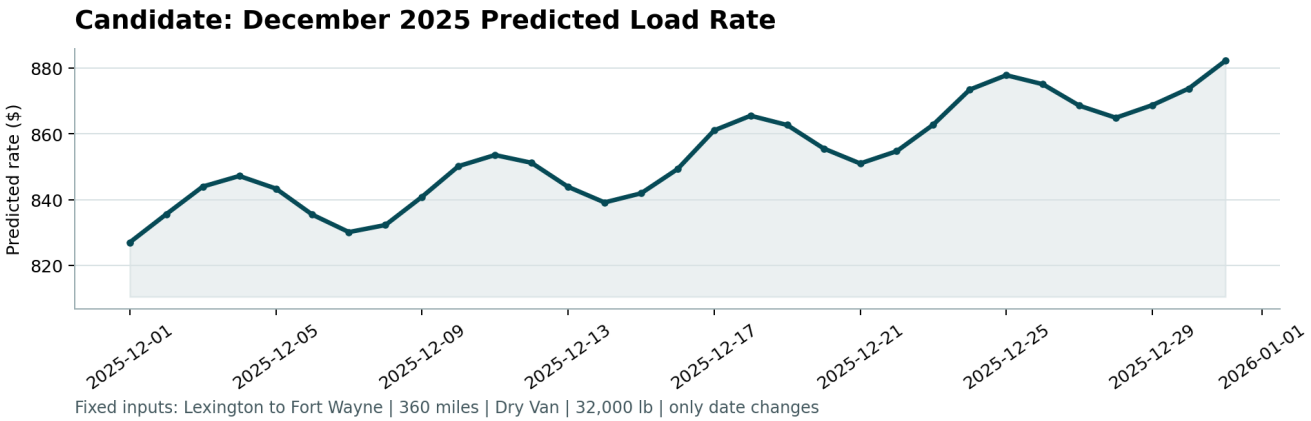


The Mar-Apr fold is the weakest because the training window (Jan-Feb) contains no quarter end yet, so the late-March surge is missed (+4% bias); after Q1 is in training the daily bias stays within about +/-2%.

4. Model

log(rate) = spline(log distance) + equipment + weight terms + premium(origin) + premium(destination) + b*market_index(day) + c*days + quarter-end spline + day-of-week, fitted with ridge on clean labels; a small LightGBM (600 trees, 15 leaves) then models the residual using location, distance, weight, equipment, market level and day-of-week (deliberately without a time index so it cannot extrapolate wrongly). Why: the data are almost exactly additive in log space, so a structured model extrapolates in time (trend, quarter-end) and to unseen cities, which tree ensembles cannot; the GBM only mops up small interactions (-7% clean MAE). Plain GBMs lose 2x in accuracy because they memorise the corrupted labels, cannot extrapolate the trend, and (with quote_signal) learn a relationship that disappears.

5. Fixed December prediction chart (produced by the provided score.py)



Predictions range from $827 (Dec 1) to $882 (Dec 31). For the 31 fixed rows, the daily market-index level is looked up for each date (from the validation file's features), which creates the weekly wiggle; the rise through the month is the quarter-end ramp plus the trend. Limitations: the quarter-end pattern is extrapolated to Q4 from three quarters; holiday effects (Thanksgiving, Christmas) cannot be learned from Jan-Oct data and are not modeled.